TidyTuesday Week 45

Pranay Gundam

Thursday 5th November, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the weekly ritual where we throw darts at the FRED database and act shocked when they occasionally stick. As always, the house rules apply: these are randomly paired economic series, nobody asked for them to be related, and any correlation you see is almost certainly an elaborate coincidence wearing a trench coat. Onward.

Monday opened with the showdown nobody requested: goods-producing employment in Springfield, Massachusetts versus a price index for proprietary and government nursing homes. The raw regression came out of the gate strutting a 0.78 R-squared and a p-value with enough zeros to make you gasp – truly a relationship for the ages. And then we detrended it, and the whole thing evaporated into a p-value of 0.52 and an R-squared of 0.02, which is statistics-speak for "please stop looking at me." Classic trend mirage: two series both drifting over the decades, pretending to hold hands, until you strip out the drift and they've never met. This is the exact species of nonsense this project exists to publicly humiliate.

Tuesday paired IT manufacturers' new orders with return on equity for Missouri banks (discontinued, rest in peace). The raw regression was significant, but at a sheepish R-squared of 0.14 it was never going to win any awards. More interesting, the detrended version actually stayed significant – weaker, sure, an R-squared of 0.06 that you'd describe as "a whisper of a relationship" – but it survived the trend-ectomy, which is more than most contestants manage. I'm not saying Missouri bank profits and national IT orders share a cosmic bond, but they at least waved at each other after the drift was removed. File under "mildly, suspiciously not-nothing."

Wednesday and Thursday both deserve a trophy, or at least a polite nod. Thursday's matchup – East North Central residential electric power prices versus food and drinking place retail sales – posted a gaudy raw R-squared of 0.74, and then held onto significance after detrending with an R-squared of 0.21. Not earth-shattering, but it didn't collapse like a folding chair, so credit where it's due. The real standout was Wednesday's, where covered employment in Puerto Rico met U.S. travel service exports: the raw regression was a flabby, barely-significant 0.06, but after detrending it got *better*, jumping to a genuinely robust 0.52 with a gorgeous p-value. A relationship that strengthens once you remove the trend is the rare and precious unicorn of this whole enterprise – the detrending giveth. Do I have a theory? Vaguely: travel, tourism, employment, islands, vibes. Is it a theory I'd defend to a journal? Absolutely not. These are random pairings and I remain legally and morally obligated to remind you that drawing conclusions here is a hobby, not a science. See you next week for more darts.

2 Date: 2026-11-02

Series ID: SMU25781000600000001
This series is titled All Employees: Goods Producing in Springfield, MA-CT (NECTA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1990-01-01 and the observation end date is 2024-12-01.The popularity of this series is 1.

Series ID: PRGONUPIHCSA
This series is titled Medical Services Expenditures by Provider: Nursing Homes: Proprietary and Government Nursing Homes Price Index and has a frequency of Annual. The units are Index 2017=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2000-01-01 and the observation end date is 2021-01-01.The popularity of this series is 1.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_PRGONUPIHCSA	**R-squared:**	0.783
Model:	OLS	**Adj. R-squared:**	0.772
Method:	Least Squares	**F-statistic:**	72.01
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	4.64e-08
Time:	21:06:51	**Log-Likelihood:**	-73.979
No. Observations:	22	**AIC:**	152.0
Df Residuals:	20	**BIC:**	154.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	186.2043	11.787	15.797	0.000	161.616	210.793
value_fred_SMU25781000600000001	-2.2652	0.267	-8.486	0.000	-2.822	-1.708

Omnibus:	5.822	**Durbin-Watson:**	0.251
Prob(Omnibus):	0.054	**Jarque-Bera (JB):**	4.103
Skew:	1.043	**Prob(JB):**	0.129
Kurtosis:	3.353	**Cond. No.**	333.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

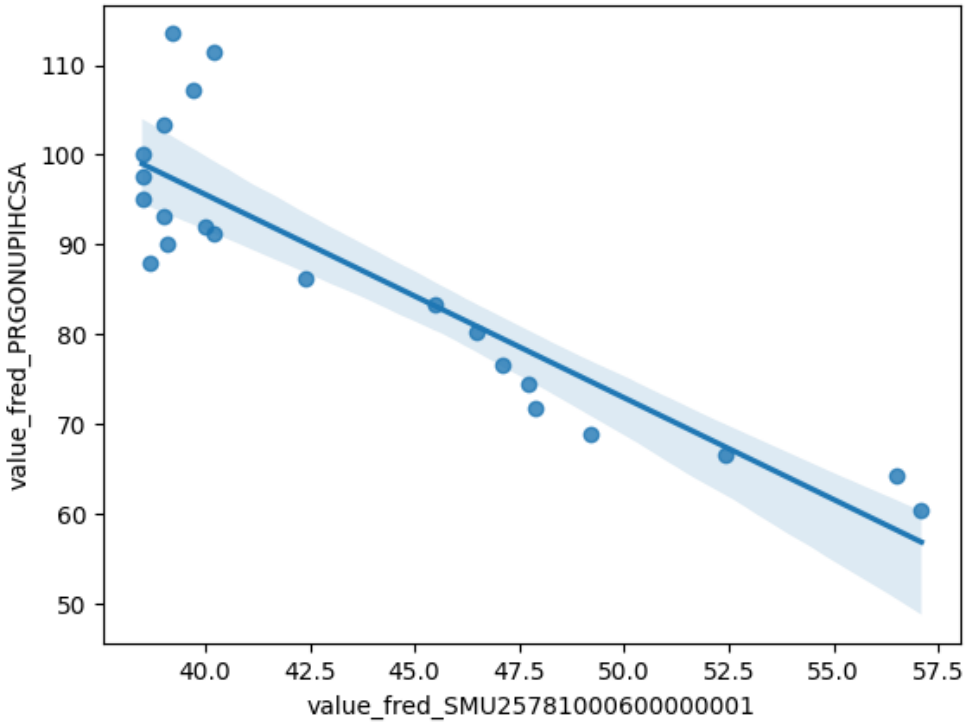


Figure 1: Raw Regression Plot for 2026-11-02

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_PRGONUPIHCSA_detrended	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	-0.028
Method:	Least Squares	**F-statistic:**	0.4223
Date:	Mon, 02 Nov 2026	**Prob (F-statistic):**	0.523
Time:	21:06:51	**Log-Likelihood:**	-42.096
No. Observations:	22	**AIC:**	88.19
Df Residuals:	20	**BIC:**	90.37
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	4.027e-13	0.367	1.1e-12	1.000	-0.765	0.765
value_fred_SMU25781000600000001_detrended	-0.0866	0.133	-0.650	0.523	-0.365	0.191

Omnibus:	2.262	**Durbin-Watson:**	0.331
Prob(Omnibus):	0.323	**Jarque-Bera (JB):**	1.135
Skew:	-0.097	**Prob(JB):**	0.567
Kurtosis:	1.904	**Cond. No.**	2.75

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

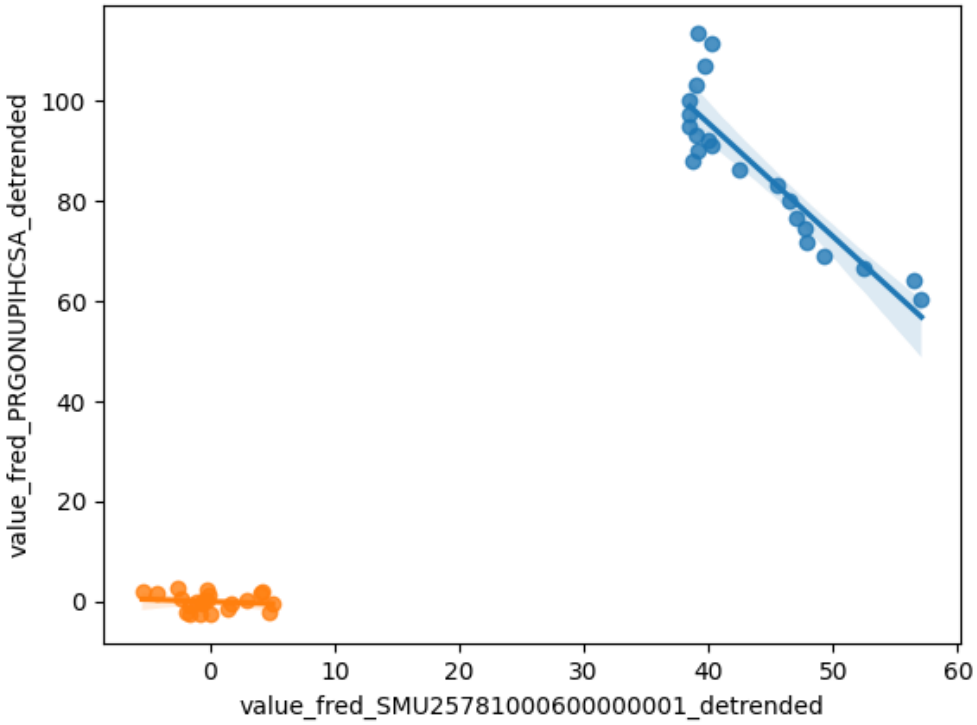


Figure 2: Detrended Regression Plot for 2026-11-02

3 Date: 2026-11-03

Series ID: AITINO
This series is titled Manufacturers' New Orders: Information Technology Industries and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-02-01 and the observation end date is 2026-09-01.The popularity of this series is 14.

Series ID: MOROE
This series is titled Return on Average Equity for Banks in Missouri (DISCONTINUED) and has a frequency of Quarterly, End of Period. The units are Percent and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1984-01-01 and the observation end date is 2020-07-01.The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MOROE	**R-squared:**	0.135
Model:	OLS	**Adj. R-squared:**	0.128
Method:	Least Squares	**F-statistic:**	17.51
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	5.69e-05
Time:	18:56:19	**Log-Likelihood:**	-307.15
No. Observations:	114	**AIC:**	618.3
Df Residuals:	112	**BIC:**	623.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.9234	2.332	0.825	0.411	-2.697	6.544
value_fred_AITINO	0.0004	9.45e-05	4.185	0.000	0.000	0.001

Omnibus:	41.602	**Durbin-Watson:**	0.161
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	88.948
Skew:	-1.485	**Prob(JB):**	4.84e-20
Kurtosis:	6.147	**Cond. No.**	1.70e+05

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.7e+05. This might indicate that there are
strong multicollinearity or other numerical problems.

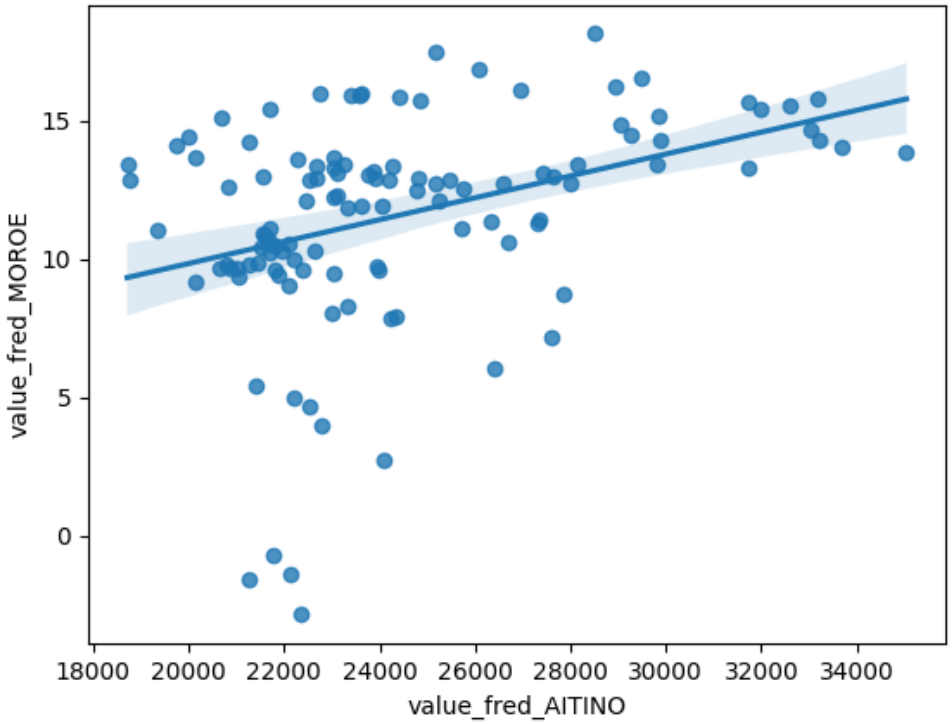


Figure 3: Raw Regression Plot for 2026-11-03

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MOROE_detrended	**R-squared:**	0.057
Model:	OLS	**Adj. R-squared:**	0.049
Method:	Least Squares	**F-statistic:**	6.779
Date:	Tue, 03 Nov 2026	**Prob (F-statistic):**	0.0105
Time:	18:56:19	**Log-Likelihood:**	-295.86
No. Observations:	114	**AIC:**	595.7
Df Residuals:	112	**BIC:**	601.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-7.674e-13	0.306	-2.5e-12	1.000	-0.607	0.607
value_fred_AITINO_detrended	0.0002	9.13e-05	2.604	0.010	5.68e-05	0.000

Omnibus:	60.278	**Durbin-Watson:**	0.190
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	192.530
Skew:	-1.974	**Prob(JB):**	1.56e-42
Kurtosis:	7.994	**Cond. No.**	3.35e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.35e+03. This might indicate that there are strong multicollinearity or other numerical problems.

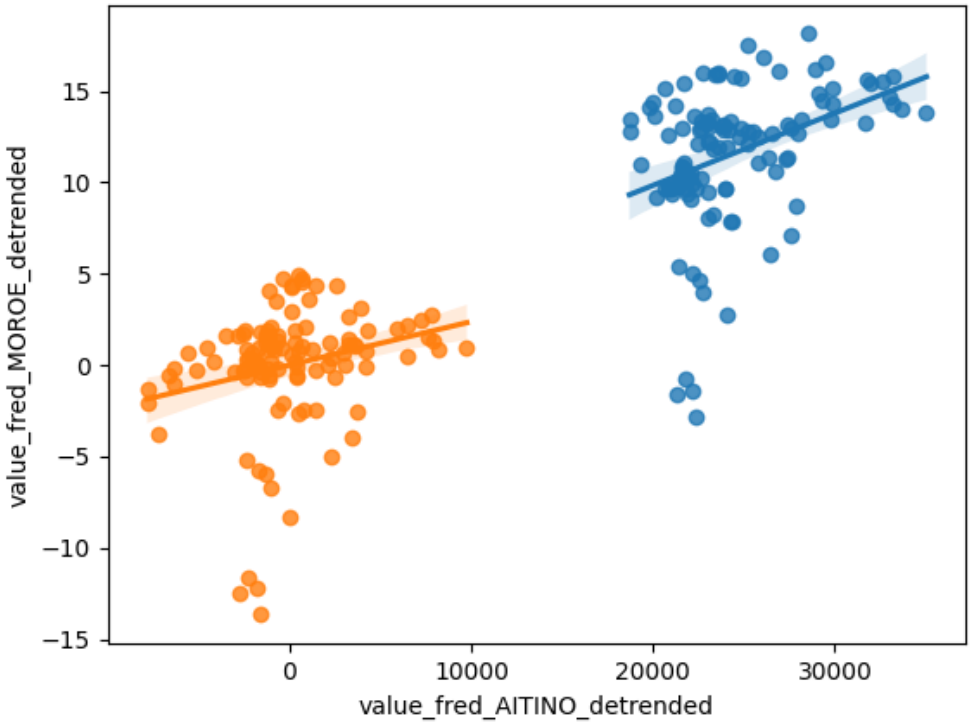


Figure 4: Detrended Regression Plot for 2026-11-03

4 Date: 2026-11-04

Series ID: WPU054121013
This series is titled Producer Price Index by Commodity: Fuels and Related Products and Power: Residential Electric Power for East North Central Census Division and has a frequency of Monthly. The units are Index Dec 2007=100 and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 2007-12-01 and the observation end date is 2026-09-01.The popularity of this series is 1.

Series ID: MRTSSM722USN
This series is titled Retail Sales: Food Services and Drinking Places and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1992-01-01 and the observation end date is 2026-08-01.The popularity of this series is 19.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MRTSSM722USN	**R-squared:**	0.737
Model:	OLS	**Adj. R-squared:**	0.734
Method:	Least Squares	**F-statistic:**	318.7
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	8.31e-35
Time:	22:07:01	**Log-Likelihood:**	-1192.6
No. Observations:	116	**AIC:**	2389.
Df Residuals:	114	**BIC:**	2395.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.164e+04	5164.973	-4.190	0.000	-3.19e+04	-1.14e+04
value_fred_WPU054121013	575.5062	32.238	17.852	0.000	511.643	639.370

Omnibus:	8.963	**Durbin-Watson:**	0.814
Prob(Omnibus):	0.011	**Jarque-Bera (JB):**	10.211
Skew:	-0.481	**Prob(JB):**	0.00606
Kurtosis:	4.089	**Cond. No.**	1.25e+03

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

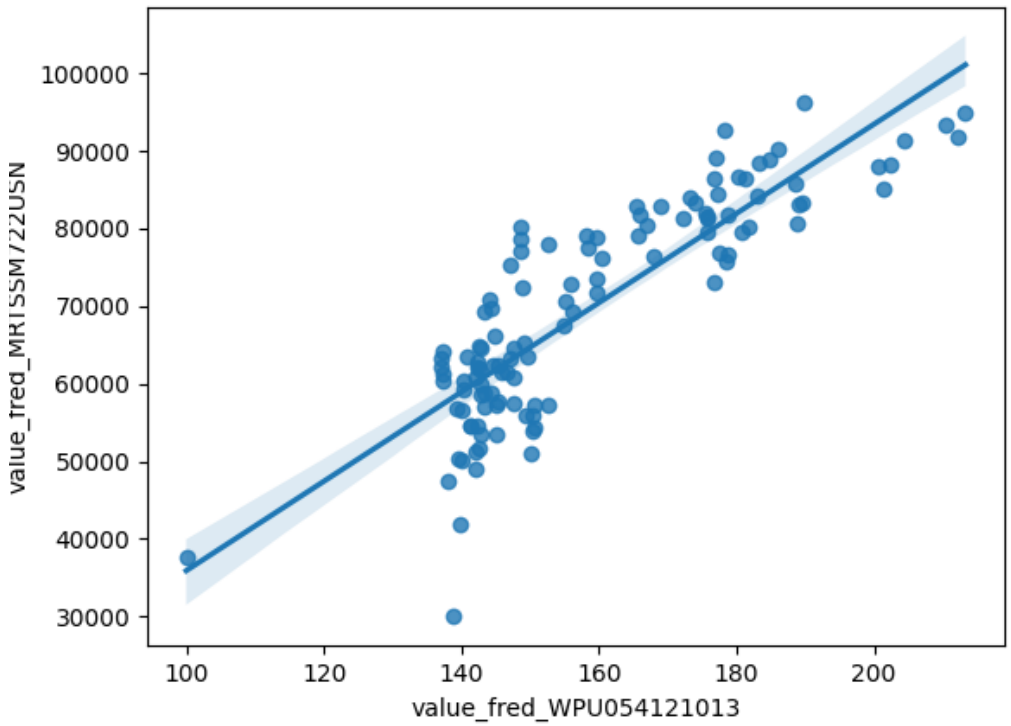


Figure 5: Raw Regression Plot for 2026-11-04

[2] The condition number is large, 1.25e+03. This might indicate that there are strong multicollinearity or other numerical problems.

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MRTSSM722USN_detrended	**R-squared:**	0.211
Model:	OLS	**Adj. R-squared:**	0.204
Method:	Least Squares	**F-statistic:**	30.56
Date:	Wed, 04 Nov 2026	**Prob (F-statistic):**	2.08e-07
Time:	22:07:01	**Log-Likelihood:**	-1180.4
No. Observations:	116	**AIC:**	2365.
Df Residuals:	114	**BIC:**	2370.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.527e-09	595.106	2.57e-12	1.000	-1178.899	1178.899
value_fred_WPU054121013_detrended	318.6043	57.635	5.528	0.000	204.429	432.779

Omnibus:	41.308	**Durbin-Watson:**	0.835
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	110.393
Skew:	-1.327	**Prob(JB):**	1.07e-24
Kurtosis:	6.975	**Cond. No.**	10.3

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

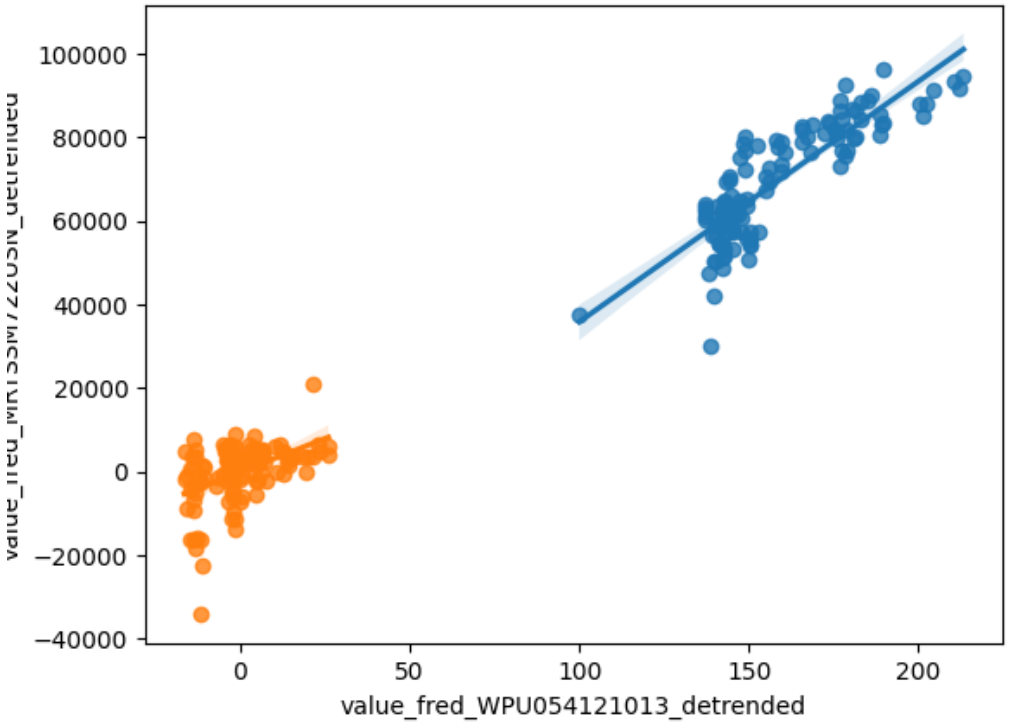


Figure 6: Detrended Regression Plot for 2026-11-04

5 Date: 2026-11-05

Series ID: PRICEMPLOY
This series is titled Covered Employment in Puerto Rico and has a frequency of Weekly, Ending Saturday. The units are Number and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1986-02-08 and the observation end date is 2026-10-17.The popularity of this series is 1.

Series ID: BOXTVLM133S
This series is titled U.S. Exports of Services - Travel and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Seasonally Adjusted.The observation start date is 1992-01-01 and the observation end date is 2017-09-01.The popularity of this series is 2.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_BOXTVLM133S	**R-squared:**	0.064
Model:	OLS	**Adj. R-squared:**	0.042
Method:	Least Squares	**F-statistic:**	2.938
Date:	Thu, 05 Nov 2026	**Prob (F-statistic):**	0.0937
Time:	19:20:26	**Log-Likelihood:**	-420.46
No. Observations:	45	**AIC:**	844.9
Df Residuals:	43	**BIC:**	848.5
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	1.907e+04	6693.471	2.850	0.007	5574.413	3.26e+04
value_fred_PRICEMPLOY	-0.0121	0.007	-1.714	0.094	-0.026	0.002

Omnibus:	5.659	**Durbin-Watson:**	0.056
Prob(Omnibus):	0.059	**Jarque-Bera (JB):**	2.550
Skew:	0.283	**Prob(JB):**	0.279
Kurtosis:	1.981	**Cond. No.**	1.51e+07

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.51e+07. This might indicate that there are
strong multicollinearity or other numerical problems.

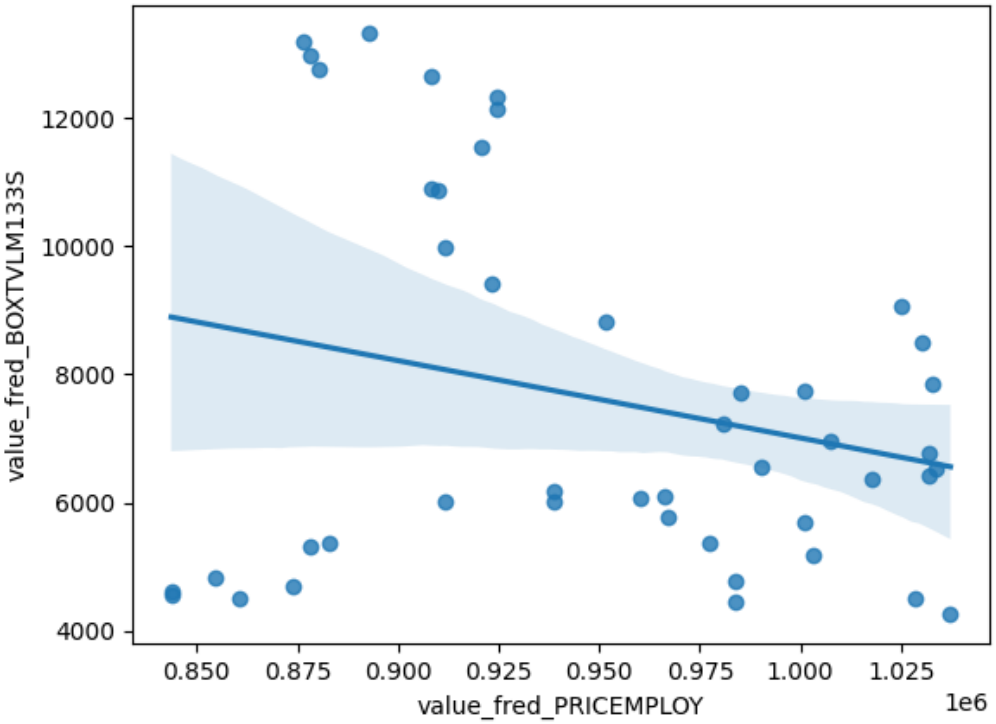


Figure 7: Raw Regression Plot for 2026-11-05

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_BOXTVLM133S_detrended	**R-squared:**	0.519
Model:	OLS	**Adj. R-squared:**	0.508
Method:	Least Squares	**F-statistic:**	46.42
Date:	Thu, 05 Nov 2026	**Prob (F-statistic):**	2.40e-08
Time:	19:20:26	**Log-Likelihood:**	-369.72
No. Observations:	45	**AIC:**	743.4
Df Residuals:	43	**BIC:**	747.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
const	-2.659e-10	136.526	-1.95e-12	1.000	-275.330	275.330
value_fred_PRICEMPLOY_detrended	-0.0156	0.002	-6.813	0.000	-0.020	-0.011

Omnibus:	0.541	**Durbin-Watson:**	0.520
Prob(Omnibus):	0.763	**Jarque-Bera (JB):**	0.636
Skew:	-0.013	**Prob(JB):**	0.727
Kurtosis:	2.418	**Cond. No.**	5.97e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 5.97e+04. This might indicate that there are strong multicollinearity or other numerical problems.

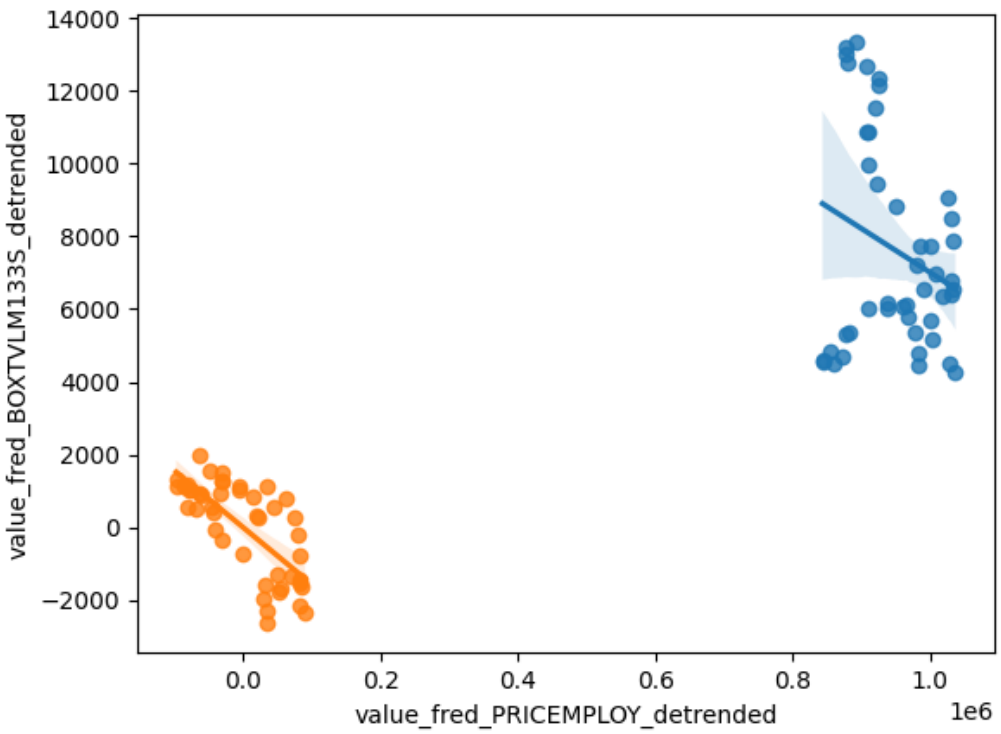


Figure 8: Detrended Regression Plot for 2026-11-05